Jenil Shingala

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Education

Bachelor of Science in Computer Science **Jan 2022 - Dec 2025**

California State University, Sacramento | Major GPA 3.8 | Dean's Honor List

- **Coursework:** Parallel & Concurrent Programming, Machine Learning, Artificial Intelligence, Operating Systems, Computer Architecture, Computer Networks, Data Structures & Algorithms
- **Certifications:** NVIDIA-Certified Associate: AI Infrastructure & Operations (NCA-AIIO) (NCCL, BlueField DPU, TensorRT, RoCE/InfiniBand, MIG, vGPU) LINK; NVIDIA-Certified Associate: Generative AI LLMs (NCA-GENL) LINK

Experience

AI Software Engineer Intern (Project Lead) **May 2025 - Dec 2025**

Advanced Integration & Controls | Sacramento, CA

- Led design and deployment of a **production agentic RAG system**, grounding LLM responses in verified documents and improving accuracy by **96%** through **AI infrastructure** optimization.
- Built a **retrieval pipeline** using **Azure AI Search**, **vector embeddings**, and **Blob Storage** for high-performance semantic search and inference, raising daily productivity by **87%** for a team of **10-25 engineers**.
- Built **API-connected automations** (Bill of Materials, PCN, Monday.com tools) across **5+ data sources**, including an **MCP integration** to **Monday.com** and **Power Automate** flows.

ML/Software Engineer **Sep 2024 - May 2025**

SentrySight LLC | Sacramento, CA

- Deployed a full-stack AI app with a **React** frontend and **AWS** backend (RDS, LightSail), serving real-time **computer-vision inference** at **<50ms latency**.
- Built a two-model deep-learning pipeline (**YOLOv11n** + transfer-learned **CNN**), achieving **91% CNN accuracy** and **94% face-recognition accuracy** across a **500+ person database**.
- Owned the full lifecycle from design through **CI/CD** deployment, building a **QA framework** with **15+ automated checks** and human-in-the-loop review to prevent quality drift before release.

Embedded Engineer Intern **Aug 2021 - Dec 2021**

Goldfield Weighting Solution

- Designed and shipped **C++** embedded firmware for industrial weighing systems, applying **object-oriented design patterns** at the hardware-software boundary to support production deployment.
- Partnered with a cross-functional team of **4** on **hardware integration**, connecting firmware to physical scale hardware to enable reliable end-to-end operation.
- Diagnosed and resolved hardware-software defects through systematic end-to-end testing, reducing the **defect rate by 23%** ahead of release.

Projects

FlashAttention GPU Kernel (from scratch) | *Triton · PyTorch · CUDA · GPU Kernels · Online-Softmax.* GITHUB

- Implemented **FlashAttention** as a fused **Triton** GPU kernel using **online-softmax** streaming recurrence, cutting peak attention memory from $\mathbf{O(N^2)}$ to $\mathbf{O(N)}$, about **250x lower** at sequence length 8192.
- Benchmarked against PyTorch `scaled_dot_product_attention` using **CUDA-event timing** and peak-memory profiling, confirming **~17MB vs ~4.3GB** peak memory at sequence length 8192; extending with **Triton autotuning** and causal masking.

AI Hackathon 2025 @ Hornet Hacks: Ticker Skimmer | *Python · Bidirectional LSTM · RNN · NLP · Gradio.* GITHUB

- Built a stock predictor combining Reddit sentiment (**NLP**) with a **bidirectional LSTM** time-series model and a **JIT optimization engine**, served through a real-time **Gradio** UI.
- Used **5-day time-step sequences** with **Z-score normalization** and **EarlyStopping** to capture pattern reversals from both past and future context.

Network Intrusion Detection System | *Python · FCNN · CNN · Keras Tuner · TensorFlow.* GITHUB

- Built a deep-learning intrusion-detection model (**FCNN/CNN** in **TensorFlow**) reaching **97% accuracy** on imbalanced network-traffic data.
- Ran **Keras Tuner** hyperparameter search and evaluated with **precision, recall, F1**, and confusion-matrix analysis to handle class imbalance.

AI Hackathon 2024 @ Hornet Hacks: Weapon and Face Detection | *YOLOv11n · OpenCV · CNN · Inception v3 · Transfer Learning · CUDA.* GITHUB

- Built a two-model **computer-vision pipeline**: **YOLOv11n** for object detection and a custom **CNN** via **transfer learning** from Inception v3 for face recognition, achieving **94% detection accuracy**.
- Reduced false positives by **87%** with dual-model agreement, processing at **<50ms latency** across a **500+ individual database** with **91% CNN accuracy**.

Technical Skills

Deep Learning & AI	LLMs, GenAI, Agentic RAG, Deep Learning, CNN, YOLO, Computer Vision, Bidirectional LSTM/RNN, NLP, Transfer Learning, AI Infrastructure, Keras Tuner, Model Evaluation (Precision/Recall/F1)
Languages & Cloud	Python, C, C++, JavaScript, React.js, Flask, AWS (RDS, LightSail), Azure, Git, CI/CD, Agile/SDLC, System Architecture (OOP), Linux
ML Systems & GPU	PyTorch, TensorFlow, CUDA, CUDA C++, Triton, GPU Kernel Development, GPU Architecture, NCCL, Distributed Data Parallel (DDP), Tensor Parallel, FlashAttention, HPC
Distributed & Perf.	Multi-GPU, RoCE/InfiniBand, Parallel Computing, Performance Optimization, Benchmarking, Memory Optimization, Triton Autotuning, Latency Optimization, Scalability